

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Energy conservation: transport, buildings, industry - Set 1

Mark scheme

AQA spec 3.3.4.2; Energy resources; textbook pp. 225-235

Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	Calculation 1 mark for the correct answer; working does not have to be shown. 1 $(2.9 - 0.7) / 2.9 \times 100 = 75.9\%$ (accept 76%) Also accept • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. Do not credit • A bare wrong answer with no working.	1
01.2	1 mark each 1 mark for each of the 2 points below. 1 Although the larger vehicle uses more energy in total, it carries many more people / a larger load 1 So the energy per passenger (or per tonne of goods) is lower – steel wheels on rails also have low rolling resistance Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
01.3	Any 3 of 4 1 mark each for any 3 of the 4 points below, in any order, maximum 3. 1 Flexibility of routes / destinations and door-to-door convenience 1 Time / speed and reliability where public transport is infrequent 1 Low fuel tax / cheap fuel 1 Carrying goods – comfort – status Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. reliability → repeatable / gives consistent results when repeated Do not credit - Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
01.4	1 mark each 1 mark for each of the 4 points below. 1 Integrated systems (park and ride, bicycles on trains, linked timetables) make it convenient to use the most efficient mode for each part of a journey 1 So more journeys shift from cars to bulk transport / cycling 1 ATM / smart motorways use variable speed limits to prevent congestion and keep traffic flowing freely 1 Vehicles avoid stop-start driving and run nearer their optimum speed (about 56 mph), reducing fuel wasted in acceleration and idling Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-228; AQA spec 3.3.4.2.1.	
02.1	2 marks per point 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. 2 Aerodynamic shapes / fairings reduce friction with the air (air resistance) so less energy is needed to propel the vehicle, especially at speed – hydrodynamics for ships 2 Reduced mass (plastics, carbon-fibre composites, high-strength steel alloys, aluminium engine blocks, lighter batteries and magnets, rounded shapes) means less energy to accelerate and less rolling resistance, so lower fuel consumption 2 Conventional friction brakes convert kinetic energy to waste heat – regenerative braking uses a generator to convert it to electricity stored in a battery (or returned to the rail grid) which then propels the vehicle, so less fuel is needed to accelerate again Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	6

Question	Marking guidance	Marks
02.2	1 mark each 1 mark for each of the 2 points below. 1 Pneumatic tyres deform as they rotate – the continual change in shape and air movement inside generates frictional heat (more if under-inflated) 1 Steel wheels do not deform so rolling resistance is very low Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	Any 2 of 4 1 mark each for any 2 of the 4 points below, in any order, maximum 2. 1 More valves per cylinder to remove exhaust gases fully so they do not mix with fresh fuel 1 Better temperature control (thermostatic fan) 1 Electronic ignition control / regular servicing to keep spark timing correct 1 Turbochargers supplying more oxygen Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-227; AQA spec 3.3.4.2.1.	
03.1	1 mark each 1 mark for each of the 3 points below. 1 The energy used to extract raw materials, manufacture the components and assemble the vehicle (for a typical car about 1.5 years of its fuel use) 1 Reduced by using recycled materials (eg recycled aluminium needs 1/20 of the energy of new) 1 Lighter designs using less material – longer vehicle life spreading the embodied energy Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. embodied energy → all the energy used to make and transport a product	3
03.2	Any 4 of 5 1 mark each for any 4 of the 5 points below, in any order, maximum 4. 1 Use of recyclable materials wherever possible 1 Easy identification of components and their composition (code stamps) 1 Easy dismantling and separation of components / materials 1 Reusable components for new vehicles 1 Compostable materials for parts that cannot be recycled Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4

Question	Marking guidance	Marks
03.3	1 mark each 1 mark for each of the 2 points below. 1 At low speed a large proportion of the energy just runs the engine rather than moving the car 1 At high speed energy is wasted overcoming air resistance (which rises with the square of speed) – optimum for many cars about 56 mph Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	1 mark for the point below. 1 Automatic stop / start systems cutting the engine when stationary – smooth driving in higher gears avoiding sudden braking / acceleration Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 227-228; AQA spec 3.3.4.2.1.	
04.1	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 Single: $5.8 \times 20 \times 12 = 1392 \text{ W}$ 1 Argon: $1.6 \times 20 \times 12 = 384 \text{ W}$ 1 Reduction = 1008 W (accept 1010) Also accept - The correct answer however it is set out, including a different but valid method. - An answer carried forward from an earlier incorrect step (error carried forward). - An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. - The converse of any point, ie the same comparison written the other way round. - The value alone - the unit (W) is already printed on the answer line. Do not credit - A bare wrong answer with no working.	3
04.2	1 mark each 1 mark for each of the 2 points below. 1 A wider gap of trapped gas reduces conduction because gas is a poor conductor 1 If the gap is too wide convection currents start in the gas, carrying heat across – the optimum depends on the gas and window size Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
04.3	1 mark each 1 mark for each of the 2 points below. 1 Low-e coating reflects long-wavelength infrared back into the room, reducing radiation losses 1 Argon / krypton / xenon have lower thermal conductivity than air so less conduction across the gap Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. infrared → heat radiation / long-wave radiation re-emitted by the Earth	2
04.4	1 mark each 1 mark for each of the 2 points below. 1 Panes must be held apart by small spacers which cannot be long, so the gap is only about 0.2 mm 1 Used where there is not room for thick gas-filled multiple glazing (eg historic buildings) Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.5	1 mark for the point below. 1 Each extra layer saves less energy (diminishing returns) while cost rises, so beyond the optimum the cost of insulation exceeds the value of energy saved Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-231; AQA spec 3.3.4.2.2.	
05.1	2 marks per point 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. 2 Orientation: south-facing windows (northern hemisphere) maximise passive solar gains - rooms needing warmth on the sunny side - fewer / smaller windows on the north side where losses are greatest - solar screens / brise soleil prevent summer overheating 2 Low SA:V (compact shapes, terraced / joined buildings, fewer storeys of small footprint) means less external surface per unit volume through which heat is lost 2 Earth sheltering: the ground is warmer than winter air and shields the building from wind, so conduction and wind-chill losses are reduced Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes.	6

Question	Marking guidance	Marks
05.2	1 mark each 1 mark for each of the 2 points below. 1 Cement manufacture (heating limestone) is very energy intensive so concrete has high embodied energy 1 Limecrete has about half the embodied energy of concrete – rammed earth, straw bales and lime mortar are lower still Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - The converse of any point, ie the same comparison written the other way round. embodied energy → all the energy used to make and transport a product	2
05.3	1 mark each 1 mark for each of the 2 points below. 1 Warm stale air leaving and cold fresh air entering pass through the exchanger in opposite directions (counter-current) separated by thin conducting pipes / plates with a large surface area 1 Heat transfers from the outgoing to the incoming air so fresh air enters already warmed Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-232; AQA spec 3.3.4.2.2.	
06.1	1 mark each 1 mark for each of the 2 points below. 1 Occupancy sensors detect infrared / movement and switch lights or appliances off when no one is in the room 1 Programmable thermostats set different temperatures for different rooms and times so unused rooms / periods are not heated Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. infrared → heat radiation / long-wave radiation re-emitted by the Earth	2
06.2	Marks shared Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. up to 2 LEDs convert a much larger proportion of electricity to light rather than heat up to 2 Washing machines with cold / low-temperature cycles and fast spins (less drying) - dishwashers using less water so less heating - cookers with double / triple glazed doors - fridges with linear compressors - LED screens using half the power of plasma - germanium transistors Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Question	Marking guidance	Marks
06.3	Any 3 of 5 1 mark each for any 3 of the 5 points below, in any order, maximum 3. 1 Turning off lights / appliances rather than standby 1 Turning thermostats and radiator valves down / off in unused rooms 1 Only heating the water needed (kettle) 1 Showering instead of bathing – eco-cycles 1 Choosing smaller vehicles / less travel Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.4	1 mark each 1 mark for each of the 2 points below. 1 Stored hot water loses heat continuously through the tank walls and pipes 1 Heating as needed (or just before) avoids these standing losses – if storage is needed the tank should be well insulated Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 232-233; AQA spec 3.3.4.2.2.	
07.1	Calculation A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. 1 Conventional: $40 / 100 = 40\%$ 1 CHP: $(35 + 45) / 100 = 80\%$ Also accept - The correct answer however it is set out, including a different but valid method. - An answer carried forward from an earlier incorrect step (error carried forward). - An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. - The value alone - the unit (%) is already printed on the answer line. CHP → using the waste heat from generation for heating Do not credit - A bare wrong answer with no working.	2
07.2	1 mark each 1 mark for each of the 2 points below. 1 Extracting less energy as electricity leaves the cooling / condenser water at a higher temperature 1 Hotter water is more useful for district heating of homes, fish farms, greenhouses Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. CHP → using the waste heat from generation for heating	2

Question	Marking guidance	Marks
07.3	1 mark for the point below. 1 Hot water loses heat and needs pumping energy over distance – heat cannot be transported as far as electricity Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • CHP → using the waste heat from generation for heating	1
07.4	1 mark each 1 mark for each of the 3 points below. 1 Long narrow pipes give a large surface area and copper is a good thermal conductor so heat transfers quickly 1 Hot and cold fluids flowing in opposite directions maintain a temperature gradient along the whole length so more heat is transferred 1 Slower flow allows more time for heat to transfer Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
07.5	1 mark each 1 mark for each of the 2 points below. 1 One large tank has a lower surface area : volume ratio than many small ones 1 A sphere has the smallest surface area for any volume – less surface means less heat loss (with insulation of low thermal conductivity) Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 233-235; AQA spec 3.3.4.2.3.	
08.1	1 mark each 1 mark for each of the 2 points below. 1 Molten iron from the blast furnace is converted directly to steel on the same site 1 So it does not cool and need re-melting – transport energy between sites is also avoided – waste heat from one process used by another (eg Kalundborg) Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
08.2	1 mark each 1 mark for each of the 2 points below. 1 Collecting and processing small dispersed quantities can use more energy in transport than is saved 1 Mass production and large-scale mining have economies of scale – some materials (glass) take more energy to make than plastic but can be refilled many times Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. recycling → reprocessing a material to make something new Do not credit 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	2
08.3	2 marks per point 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. 2 Transmitting at high voltage means low current so less energy is lost as heat in the cables (loss proportional to current squared) – overhead cables cheaper than underground which need cooling 2 ICT / smart grids co-ordinate real-time data on demand and supply from many sources, switching generation, storage (pumped storage, batteries) and flexible loads so surplus is not wasted and inefficient peaking plants are avoided Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. pumped storage → water pumped uphill when demand is low and released to generate at peak	4
08.4	Any 2 of 4 1 mark each for any 2 of the 4 points below, in any order, maximum 2. 1 Proximity to the grid / demand to reduce transmission losses and cable costs 1 Access to cooling water for thermal stations 1 Resource availability (wind, tidal range, fuel deliveries) 1 Land availability and planning / public acceptance – stable geology Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 234-235; AQA spec 3.3.4.2.3.	